Group 1

Lab 08 Bias-Variance Tradeoff

ITAI-1371

Viswanatha Rao

Team Member Contribution Journal

Viktoriya Kurmisheva:

Worked on Step 3 of Task 1 in Part 3, which focused on making predictions using the trained polynomial regression models with degrees 1, 4, and 15. Also worked on Step 2 of Task 2 in Part 4 by plotting the learning curve for the underfitting model (degree 1) to compare its training and cross-validation scores. Answered Question 2 in the Knowledge Check section by explaining how the scores changed as the training size increased and what they showed about underfitting. After completing the lab, wrote the group's Reflection Journal based on the group's thoughts, activities, model results, graphs, and concepts covered in the notebook. Then uploaded the completed Reflection Journal to the group's GitHub repository.

Tien Manh Nguyen:

Worked on Steps 1 and 2 of Task 1 in Part 3, which involved creating the polynomial regression models with degrees 1, 4, and 15 and training them using the synthetic dataset. Also worked on Step 1 of Task 2 in Part 4 by creating the underfitting and overfitting models for the learning curve comparison. Answered Question 1 in the Knowledge Check section by identifying which polynomial models showed underfitting, overfitting, and a better fit based on their graphs. Adjusted the notebook's Markdown formatting to make the instructions, headings, and answers easier to read. After the group completed the tasks, reviewed the notebook to check the code, graphs, and answers for accuracy and consistency. Then uploaded the completed Jupyter Notebook to the group's GitHub repository.

Zaid Alayedy:

Worked on Step 4 of Task 1 in Part 3, which focused on plotting the predictions for the polynomial regression models with degrees 1, 4, and 15 to compare their complexity and how closely they followed the data. Also worked on Step 3 of Task 2 in Part 4 by plotting the learning curve for the overfitting model (degree 15) to compare its training and cross-validation scores. Answered Question 3 in the Knowledge Check section by explaining the gap between the training and cross-validation scores and what it showed about overfitting. In addition, gathered the contribution details from each group member and used them to write and organize the Team Member Contribution Journal. Reviewed the descriptions to make sure the journal clearly explained everyone's responsibilities.